

WHALES, PORPOISES, AND DOLPHINS

Collectively known as cetaceans, this group of mammals is wholly aquatic and all but six species in the order are found in coastal and marine waters.

Supremely adapted to life in water, the cetacean body is tapered and streamlined, with the forelimbs modified into flippers. There are no visible hind limbs, but the tail has horizontal flukes for added propulsion. Many species also have a dorsal fin. The skin is almost hairless and the body is insulated by a layer of blubber, which is especially thick in species living in cold water.

BREATHING AND COMMUNICATING

Capable of diving to great depths and for long periods because of their ability to store oxygen in their muscle tissues, cetaceans must come to the surface to breathe. Breathing is through nostrils, known as blowholes, which are located on the top of the head. When stale air is expelled, it may create a spout of condensation – the spout's size, angle, and shape can be used to distinguish some species, even when their body is almost fully submerged.

Most species produce sound. Some use a series of clicks for echolocation. The clicks bounce off nearby objects, alerting them to any obstacles on their path. Others communicate with each other using vocalizations that range from whistles and groans to the complex songs of many great whales. While their hearing is good, the ears are reduced to simple openings behind the eyes. External ears are both undesirable – they would affect streamlining – and unnecessary, since water is an effective medium for sound transmission.

HUNTER OR FILTER-FEEDER

Cetaceans are divided into two main groups based on their feeding habits. Toothed species, predators of fish, large invertebrates, seabirds, seals, and sometimes smaller cetaceans, catch prey with their sharp teeth and usually swallow it whole, without chewing. In contrast, filter-feeding baleen whales have fibrous baleen plates suspended from their upper jaw that act like sieves. Water, containing invertebrates and small fish, is taken in and then forced out through the plates by the tongue, leaving behind the organisms.

PHYLUM	CHORDATA
CLASS	MAMMALIA
ORDER	CETACEA
FAMILIES	14
SPECIES	About 90

DEBATE

LAND-LIVING ANCESTORS

The taxonomic position of cetaceans has long been debated; their extreme adaptation to a wholly aquatic lifestyle masks anatomical features shared with other orders. Today it is generally accepted that Cetacea is most closely related to Artiodactyla – specifically the hippopotamus family – and so they are grouped together as Cetartiodactyla, at order or superorder level. Evidence comes mainly from genetic and molecular studies, but there is increasing anatomical support, including strong similarities between the ankle bones of artiodactyls and some fossil ancestors of whales.

RIGHT WHALES

Found in cool temperate and polar waters, the Balaenidae were considered the “right” ones to hunt, as they are easily approached, often close to shore, and have a thick layer of oily blubber – an adaptation to their cool water habitats. Right whales and the bowhead whale lack dorsal fins and throat grooves and their strongly curved jaws support the longest baleen plates of any whale.

SOUTHERN RIGHT WHALE *Eubalaena australis*

Restricted to the southern hemisphere, this species has distinctive areas of pale hard skin (callosities) on its large head. These are used to identify individuals.

irregular white patches on belly

15–16.5 m
50–54 ft

thick tail stock

rounded back,
with no dorsal fin

pale area at
base of tail

20 m
65 ft

broad, pointed
tail flukes

BOWHEAD WHALE *Balaena mysticetus*

Named after its arching lower jaw, this robust whale has very long baleen plates. It spends its whole life in Arctic waters, feeding on plankton, which thrive in the cold, oxygen-rich waters. When threatened, the bowhead retreats under the ice.

irregular white
chin patch

blowhole

